

Confidence-Gated Structured Reasoning for Zero-Shot Meme Hate Detection

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Abstract

Hateful memes are difficult to detect because their meaning often emerges from the interaction between image and text rather than from either modality alone. Recent vision-language models (VLMs) can classify many memes in a zero-shot setting, but it is unclear when additional structured reasoning is useful and when it simply adds cost or introduces new errors. We first compare three recent open-source VLMs on a 3000-example subset of the Hateful Memes benchmark, where Qwen3-VL-8B achieves the highest Accuracy. We then use this model in a confidence-gated refinement pipeline that applies additional reasoning only to low-confidence predictions. We compare four refinement strategies: Atomic Fact Extraction, Static Question Answering, Dynamic Question Answering, and Chain-of-Thought reasoning. Atomic Fact Extraction performs best, improving Accuracy from 0.7177 to 0.7230 and Macro F1 from 0.7027 to 0.7109. Its advantage appears to lie in separating observable facts from the final judgment of hatefulness. We also find that more reasoning is not always better, as some strategies increase hateful-class recall while reducing overall accuracy. These findings suggest that selective refinement can support multimodal hate detection, but the choice of reasoning strategy is crucial.

1 Introduction

Online hateful content remains a growing concern for social media platforms and online communities. Hateful memes are particularly difficult to moderate because the hateful meaning often emerges from the interaction between image and text rather than from either modality independently. A caption that appears harmless on its own may become offensive when paired with a specific image, while an image that appears benign may acquire hateful meaning through accompanying text. Humour,

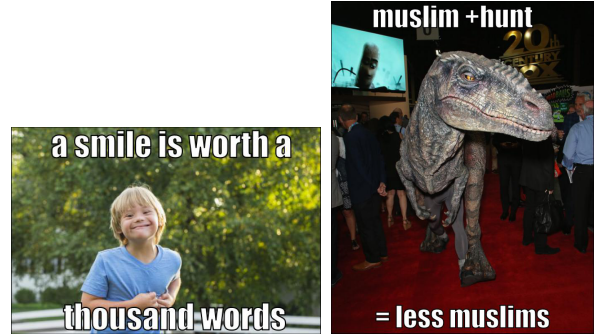


Figure 1: Examples from the Hateful Memes dataset. Left shows a non-hateful meme. Right shows a hateful meme where the combined image-text context promotes hostility toward a protected religious group. The right example contains offensive content and is included only to illustrate the task.

sarcasm, stereotypes, and coded language further increase the difficulty of automated moderation systems.

Figure 1 shows examples from the Hateful Memes benchmark (Kiela et al., 2020). The meme on the left is non-hateful because the image and text together express a harmless motivational message. In contrast, the meme on the right combines the image and caption to promote hostility toward a protected religious group. These examples illustrate that hateful meme detection requires joint multimodal reasoning over both visual and textual information.

To study this multimodal challenge, Kiela et al. (2020) introduced the Hateful Memes benchmark, which was specifically designed to prevent unimodal shortcuts. The dataset includes benign confounders where either the image or the text from a hateful meme is reused in a different non-hateful context. This forces models to understand how image and text interact instead of relying only on offensive keywords or suspicious visual patterns.

Existing hateful meme detection approaches often rely on supervised multimodal fusion archi-

tures or fine-tuned multimodal transformers (Gómez et al., 2020; Zhu et al., 2022). While these approaches can improve benchmark performance, they also introduce limitations. Labelled hateful meme datasets remain limited in size and may not adequately cover emerging meme formats, unseen target groups, or evolving stereotypes. Models trained on specific hateful categories may therefore learn dataset-specific patterns instead of more general multimodal reasoning behaviour. For this reason, zero-shot evaluation is important because it evaluates whether models can generalise to unseen hateful contexts without additional task-specific training (Zhu et al., 2022).

Recent large vision-language models have made zero-shot hateful meme detection increasingly practical. Prior work shows that direct prompting already provides a strong baseline for the task (Yamagishi, 2024). Other work suggests that structured reasoning can improve multimodal understanding by encouraging the model to reason more explicitly about the meme target, stereotype, and image-text interaction (Nandi et al., 2024). However, these approaches typically apply structured reasoning uniformly across all examples.

Applying long reasoning pipelines to every meme introduces two important limitations. First, structured reasoning substantially increases inference cost and processing time. Second, forcing a model to deeply analyse simple examples may cause the model to over-reason and infer hateful intent that does not actually exist. In some cases, an initially correct prediction may become incorrect after unnecessary refinement.

In practice, not all memes require the same level of reasoning. Many examples can already be classified correctly using a simple direct prediction, while only a smaller subset of difficult examples may benefit from additional reasoning. Motivated by this observation, we investigate whether structured reasoning should be triggered selectively rather than uniformly. Similar confidence-guided refinement strategies have recently been explored in reasoning-based NLP systems (Jang et al., 2025).

In this work, we propose a confidence-gated structured reasoning pipeline for zero-shot hateful meme detection.¹ The system first generates a direct prediction together with a confidence estimate derived from label likelihood probabilities com-

puted from the model outputs. Only predictions below a predefined confidence threshold are routed to an additional refinement stage. We first compare multiple open-source vision-language models for the initial prediction stage and then evaluate four refinement strategies, including Atomic Fact Extraction, Static QA, Dynamic QA, and Chain-of-Thought reasoning, using the best-performing model in a strict zero-shot setting.

Our contributions are as follows:

- We propose a confidence-gated structured reasoning pipeline for zero-shot hateful meme detection.
- We investigate whether selective refinement can reduce unnecessary reasoning on high-confidence examples.
- We compare four structured refinement strategies including Atomic Fact Extraction, Static QA, Dynamic QA, and Chain-of-Thought reasoning.
- We analyse how different confidence thresholds affect refinement frequency and downstream classification performance.

We address three research questions:

- Does a confidence-gated zero-shot structured reasoning pipeline outperform a direct prompting baseline on the Hateful Memes benchmark?
- Which refinement strategy among Atomic Fact Extraction, Dynamic QA, Static QA, and Chain-of-Thought reasoning is most effective for improving low-confidence predictions?
- How does the confidence threshold influence the trade-off between computational efficiency and classification performance?

2 Related Work

Multimodal Hate Meme Detection Hateful meme detection has emerged as a challenging multimodal classification task because the hateful meaning often depends on the interaction between image and text rather than either modality independently. To study this problem, Kiela et al. (2020) introduced the Hateful Memes benchmark, which includes benign confounders designed to prevent models from relying on unimodal shortcuts. In many cases, the same image or text may appear in both hateful and non-hateful contexts, forcing models to jointly reason over both modalities.

Early approaches to hateful meme detection primarily relied on supervised multimodal fusion architectures that combined textual and visual repre-

¹Code available at: <https://github.com/deepaksoresh/Confidence-Gated-Q-A-Reasoning-for-Zero-Shot-Meme-Hate-Detection>

sentations (Gómez et al., 2020; Zhu et al., 2022). Several works also explored fine-tuning large multi-modal transformers directly on the Hateful Memes dataset. While these approaches improved benchmark performance, they typically depended on labelled hateful meme datasets and often struggled to generalise to unseen meme formats, target groups, or evolving stereotypes. Recent work has therefore emphasised the importance of studying hateful meme detection in more generalizable settings such as zero-shot evaluation (Zhu et al., 2022).

Zero-shot Vision-Language Models Recent advances in large vision-language models have significantly improved zero-shot multimodal reasoning capabilities. Models such as LLaVA (Liu et al., 2023) and Qwen-VL (Qwen Team, 2025) demonstrate strong performance across a wide range of visual reasoning and image understanding tasks without requiring task-specific fine-tuning.

These capabilities have also been explored for hateful meme detection. Yamagishi (2024) showed that direct prompting with large vision-language models already provides a strong zero-shot baseline on the Hateful Memes benchmark. Their findings suggest that simple prompting strategies can outperform more complicated prompting formulations in some settings. This indicates that modern vision-language models already possess substantial multimodal reasoning knowledge that can be transferred to hateful meme classification without additional supervised training.

Structured Reasoning and Confidence-Guided Refinement Several recent studies have explored whether structured reasoning can further improve multimodal understanding. Nandi et al. (2024) proposed SAFE-MEME, a structured reasoning framework that guides the model through intermediate question-answering steps before producing a final hateful meme prediction. By explicitly reasoning about targets, stereotypes, and image-text interactions, structured prompting can reduce some classification errors compared to direct prediction alone.

More broadly, reasoning-based prompting strategies such as Chain-of-Thought prompting (Wei et al., 2022) and decomposition-based reasoning frameworks (Stacey et al., 2024) have shown improvements across multiple language and vision-language tasks. However, these approaches generally apply additional reasoning uniformly across all inputs, regardless of difficulty or model confidence.

Recent work in NLP has started exploring

confidence-guided refinement strategies to address this limitation. Jang et al. (2025) proposed Confidence-guided Refinement Reasoning (C2R), where additional reasoning is triggered only for low-confidence predictions. This selective routing strategy reduces unnecessary reasoning while maintaining strong downstream performance.

Our work bridges these directions by combining confidence-guided routing with structured multimodal reasoning for zero-shot hateful meme detection. Unlike prior structured reasoning approaches that apply refinement to all examples, our framework selectively triggers additional reasoning only when the initial prediction confidence falls below a predefined threshold. We further compare multiple structured refinement strategies, including Atomic Fact Extraction, Static QA, Dynamic QA, and Chain-of-Thought reasoning within the same zero-shot evaluation setting.

3 Data

Data We use a publicly available version of the Hateful Memes benchmark introduced by Kiela et al. (2020). The dataset contains meme images, text extracted from each meme, and a binary label indicating whether the meme is hateful or non-hateful. The benchmark was designed as a multi-modal classification task where hateful meaning often depends on the interaction between image and text.

Evaluation subset The original benchmark contains 10,000 examples. Due to the computational cost of repeated multi-stage refinement experiments across multiple confidence thresholds and reasoning strategies, we evaluate all methods on a fixed subset of 3000 English examples. The subset contains 1064 hateful memes and 1936 non-hateful memes, with no duplicate IDs. The same subset is used consistently across all experiments to ensure fair comparison between models and refinement strategies.

Annotation We use the original labels provided with the dataset and do not perform additional annotation.

4 Methodology

4.1 Pipeline Overview

Our framework consists of three stages, as illustrated in Figure 2. In the first stage, we evaluate multiple open-source vision-language mod-

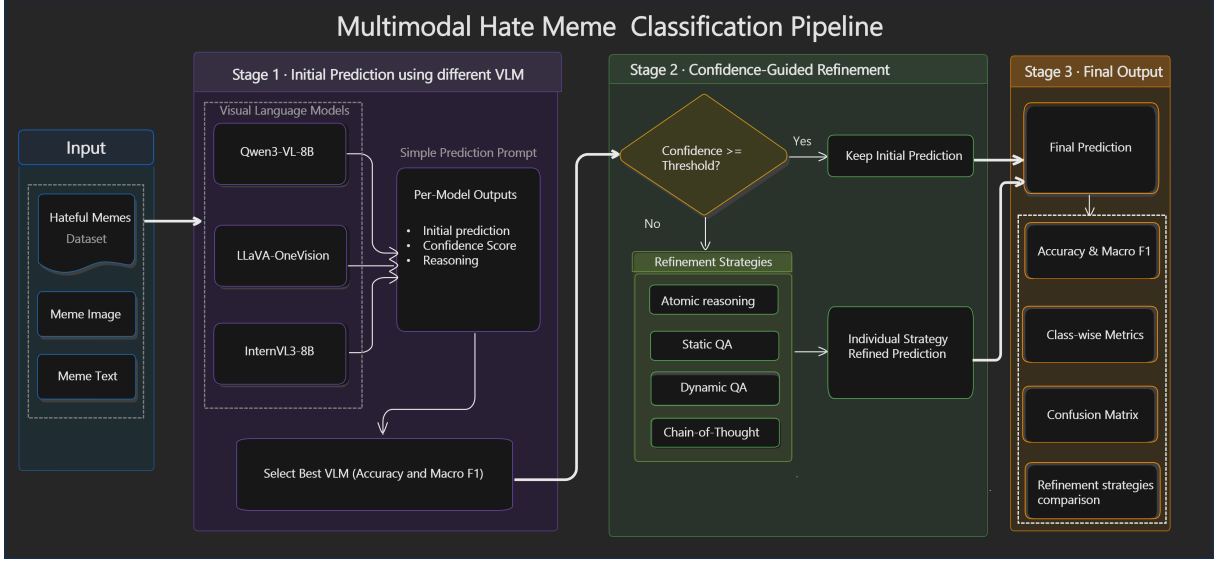


Figure 2: Overview of the proposed confidence-gated structured reasoning pipeline for zero-shot hateful meme detection. Multiple vision-language models are first evaluated using direct prompting. The selected model is then used for confidence-gated refinement experiments. Predictions with confidence greater than or equal to a predefined threshold are accepted directly, while low-confidence predictions are routed to one of four refinement strategies, including Atomic Fact Extraction, Static QA, Dynamic QA, and Chain-of-Thought reasoning.

els using a direct zero-shot prompting setup for hateful meme classification. Specifically, we compare Qwen3-VL-8B-Instruct (Qwen Team, 2025), LLaVA-OneVision-Qwen2-7B (Li et al., 2024), and InternVL3-8B (Zhu et al., 2025). We focus on a zero-shot setting to evaluate whether modern vision-language models can generalise to hateful meme understanding without task-specific fine-tuning.

Each model receives the meme image together with the extracted meme text and produces a binary hateful or non-hateful prediction, a confidence score, and a short textual explanation. We then compare the models using Accuracy and Macro F1 score and select the best-performing vision-language model for the refinement stage. Comparing multiple models allows us to analyse whether confidence-guided refinement depends on the underlying reasoning capability of the base model.

In the second stage, we apply confidence-gated structured reasoning using the selected model. For each meme, the system first generates an initial prediction together with a confidence estimate derived from label likelihood probabilities. Predictions with confidence greater than or equal to a predefined threshold are accepted directly as the final output. Predictions below the threshold are routed to an additional refinement stage. This design is motivated by the observation that not all memes

require the same level of reasoning complexity and that unnecessary refinement may introduce over-reasoning errors.

We investigate four refinement strategies, including Atomic Fact Extraction, Static QA, Dynamic QA, and Chain-of-Thought reasoning. Each strategy encourages the model to reason about the meme in a different way before producing a refined verdict. The refinement strategies were selected to compare both constrained and unconstrained forms of structured multimodal reasoning. We further evaluate multiple confidence thresholds to analyse the trade-off between refinement frequency and downstream classification performance.

In the final stage, the predictions from the confidence-gated pipeline are aggregated and evaluated using standard classification metrics. We additionally compare the refinement strategies across different confidence thresholds to analyse their effectiveness under varying levels of model uncertainty.

4.2 Initial Prediction

For the initial prediction stage, we use a direct zero-shot prompting setup similar to Yamagishi (2024). Each model receives the meme image together with the extracted meme text and produces a binary hateful or non-hateful prediction. We use a simple prompting strategy because prior work has shown that overly complex prompts can sometimes

reduce zero-shot performance in multimodal hate speech detection.

Initial Prediction Prompt

Does this image contain hate speech?
Text in image: {meme_text}
Answer with ‘Yes’ or ‘No’ and provide a brief reason.

A prediction of ‘Yes’ corresponds to the hateful class, while ‘No’ corresponds to the non-hateful class. The generated response is parsed into a binary prediction by checking whether the response starts with ‘Yes’ or ‘No’. The textual explanation produced by the model is stored for later qualitative analysis but is not used for prediction selection or confidence estimation.

4.3 Confidence Estimation

A central component of our framework is the confidence estimation mechanism used for routing predictions to the refinement stage. Following common practice in confidence calibration, we treat the normalised probability assigned to the predicted label as the model confidence (Guo et al., 2017). Instead of relying on self-reported confidence, we estimate this value directly from the model likelihood assigned to each candidate label.

For each example, we score the candidate labels *Yes* and *No*. Each candidate label is appended to the same rendered prompt, and the model loss is calculated only over the appended label tokens. The prompt tokens are masked using the value -100, so they do not contribute to the loss calculation.

For each label y_i , we convert the label loss into a score:

$$s_i = -\mathcal{L}(y_i) \cdot n_i \quad (1)$$

where $\mathcal{L}(y_i)$ represents the cross-entropy loss for candidate label y_i , and n_i denotes the number of tokens in that candidate label. The resulting scores for *Yes* and *No* are normalised using a softmax:

$$P(y_i | x) = \frac{\exp(s_i)}{\sum_j \exp(s_j)} \quad (2)$$

The final prediction is obtained from the generated answer, while the confidence score is defined as the normalised probability assigned to that predicted label. This produces a confidence estimate between 0 and 1 that is later used by the confidence gate for selective refinement.

4.4 Confidence-Gated Refinement

Based on the initial model comparison results, the selected vision-language model is used for the refinement experiments. The saved initial predictions and confidence scores from this model are reused to determine whether each example should undergo additional refinement.

Let $c(x)$ denote the confidence score assigned to the initial prediction for example x , and let t denote the confidence threshold. If $c(x) \geq t$, the initial prediction is accepted as the final prediction. If $c(x) < t$, the example is routed to the refinement stage.

$$\hat{y}_{final} = \begin{cases} \hat{y}_{initial}, & \text{if } c(x) \geq t \\ \hat{y}_{refined}, & \text{if } c(x) < t \end{cases} \quad (3)$$

This routing mechanism avoids applying additional reasoning to examples that the model already classifies with high confidence. At the same time, uncertain examples are given an additional opportunity for structured reasoning. We evaluate four thresholds $t \in \{0.60, 0.75, 0.85, 0.95\}$; to avoid redundant computation, refinement is run once per technique for all examples below $t = 0.95$ and reused across lower thresholds.

4.5 Refinement Strategies

For predictions below the confidence threshold, the selected model is routed to an additional refinement stage. We compare four refinement strategies that encourage different forms of multimodal reasoning before producing a final verdict. Each strategy receives the meme image together with the meme text and must end with a structured output in the format FINAL VERDICT: Yes or FINAL VERDICT: No. This standardised format allows consistent parsing across all refinement methods.

4.5.1 Atomic Fact Extraction

The Atomic Fact Extraction strategy asks the model to separate observation from judgment. The model first extracts objective visual or textual facts from the meme, then explains how these facts interact, and finally decides whether the meme expresses hostility toward a protected group. This strategy is motivated by atomic decomposition methods, which break an input into smaller factual units before making a final prediction (Stacey et al., 2024). We include this strategy because direct reasoning over the full meme may cause the model to miss small but important visual or textual cues, while

fact extraction encourages the model to explicitly identify the evidence before making a decision.

Atomic Fact Extraction Prompt

Step 1: Extract up to 3 objective visual or textual facts from the meme.
 Step 2: Explain how these facts interact with each other.
 Step 3: Decide whether the meme targets a protected group with hostility, slurs, dehumanisation, or harmful stereotypes.
 End with exactly one line:
 FINAL VERDICT: Yes
 or
 FINAL VERDICT: No

4.5.2 Static Question Answering

The Static Question Answering strategy uses the same predefined reasoning questions for every meme. The questions focus on identifying the target, checking the image-text interaction, and deciding whether the target is a protected group. This strategy follows the idea that structured intermediate questions can guide the model toward more explicit multimodal reasoning (Nandi et al., 2024). We include this strategy to test whether a fixed reasoning template is sufficient for low-confidence examples.

Static QA Prompt

Q1. Who or what is the main target of the meme?
 Q2. Does the image-text combination express hostility, ridicule, dehumanisation, or a harmful stereotype?
 Q3. Is the target a protected group such as race, religion, gender, sexual orientation, disability, nationality, or a similar identity group?
 End with exactly one line:
 FINAL VERDICT: Yes
 or
 FINAL VERDICT: No

4.5.3 Dynamic Question Answering

The Dynamic Question Answering strategy allows the model to generate its own intermediate questions before answering them. Unlike Static QA, the reasoning questions are adapted to the specific meme. This allows the model to focus on the most relevant target, stereotype, or image-text relation for each example. We include this strategy to test

whether adaptive question generation is more useful than a fixed question template.

Dynamic QA Prompt

Generate 3 useful questions that help decide whether this meme contains hate speech. The questions should focus on the target, the image-text interaction, and whether a protected group is attacked through hostility, slurs, dehumanisation, or stereotypes.
 Answer each question based only on the meme image and text. Then make a final decision.
 End with exactly one line:
 FINAL VERDICT: Yes
 or
 FINAL VERDICT: No

4.5.4 Chain-of-Thought Reasoning

The Chain-of-Thought strategy asks the model to reason step by step before producing a final prediction (Wei et al., 2022). Compared with the QA-based methods, this strategy is less constrained because the model is not forced to answer predefined questions. We include it as a general reasoning baseline against the more structured refinement strategies.

Chain-of-Thought Prompt

Decide whether this meme contains hate speech. Consider the image and text together, identify the target, and check whether the meme attacks a protected group through hostility, slurs, dehumanisation, or harmful stereotypes.
 End with exactly one line:
 FINAL VERDICT: Yes
 or
 FINAL VERDICT: No

4.6 Final Prediction Parsing

Each refinement prompt requires the model to end with FINAL VERDICT: Yes or FINAL VERDICT: No. The final label is parsed from this line, where ‘Yes’ corresponds to hateful and ‘No’ corresponds to non-hateful. If no valid verdict is found, the system keeps the initial prediction.

5 Experimental Setup

Models We evaluate three recent open-source vision-language models from different model families: Qwen3-VL-8B-Instruct (Qwen Team, 2025), LLaVA-OneVision-Qwen2-7B (Li et al., 2024),

Model	Acc.	Macro F1	H-F1	H-Rec.
Qwen3-VL-8B	0.7177	0.7027	0.6360	0.6955
InternVL3-8B	0.7057	0.7031	0.6755	0.8637
LLaVA-OV-7B	0.6497	0.6366	0.5677	0.6485

Table 1: Initial zero-shot model comparison on the 3000-example evaluation subset. H-F1 and H-Rec. denote hateful-class F1 and hateful-class recall.

and InternVL3-8B (Zhu et al., 2025). These models were selected because they are comparable in scale and represent different VLM design families. All experiments are performed without fine-tuning or parameter updates. Based on the initial comparison results, Qwen3-VL-8B-Instruct is selected for the confidence-gated refinement experiments.

Implementation Details All experiments are implemented using Python, PyTorch, Hugging Face Transformers, and scikit-learn. We use greedy decoding with `do_sample=False`, so generation is deterministic and no sampling seed is required. The maximum generation length is set to 60 tokens for the initial prediction stage and 350 tokens for the refinement stage. Experiments were conducted on the LIACS ALICE high-performance computing cluster using GPU nodes with approximately 25GB of GPU memory.

Confidence Thresholds We evaluate four confidence thresholds, $t \in \{0.60, 0.75, 0.85, 0.95\}$. Lower thresholds route only highly uncertain examples to refinement, while higher thresholds trigger refinement more aggressively.

Evaluation Metrics We report Accuracy, Balanced Accuracy, Macro Precision, Macro Recall, and Macro F1. We also report class-wise Precision, Recall, and F1 for hateful and non-hateful classes. For the confidence-gated experiments, we additionally report refined examples, refinement rate, changed predictions, and net correction gain.

6 Results

6.1 Initial Model Comparison

Table 1 presents the initial zero-shot performance of the evaluated vision-language models on the 3000-example evaluation subset. All models are evaluated without task-specific fine-tuning or parameter updates. Qwen3-VL-8B achieves the highest overall accuracy and is therefore selected for the confidence-gated refinement experiments, while InternVL3-8B achieves the highest hateful-class F1

Strategy	Thr.	Acc.	F1	Ref.	Gain
Atomic	0.95	0.7230	0.7109	653	16
CoT	0.60	0.7187	0.7049	371	3
Dynamic QA	0.85	0.7173	0.7063	495	-1
Static QA	0.75	0.7150	0.7070	426	-8

Table 2: Best confidence-gated refinement result for each strategy. F1 denotes Macro F1, Ref. denotes refined examples, and Gain denotes net correction gain over the initial baseline.

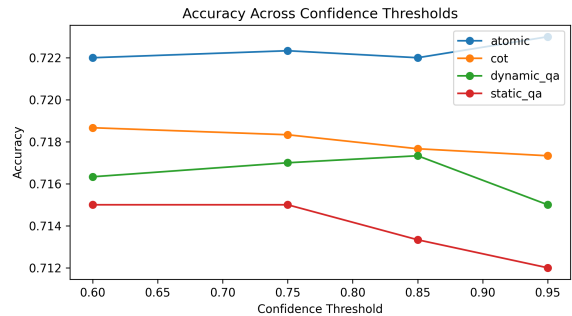


Figure 3: Accuracy across confidence thresholds for each refinement strategy.

and recall. We prioritise accuracy as the selection criterion because the class imbalance in the dataset (65% non-hateful) reflects realistic real-world moderation distributions, where non-hateful content predominates and overall system reliability matters alongside hateful-class detection. Qualitative examples of correct and incorrect high-confidence predictions are provided in Appendices A and B.

6.2 Confidence-Gated Refinement Results

Table 2 summarises the best-performing configuration for each refinement strategy across the evaluated confidence thresholds. Atomic Fact Extraction achieves the strongest overall result, improving the initial prediction baseline from 0.7177 to 0.7230 accuracy and from 0.7027 to 0.7109 Macro F1. It also produces the highest positive net correction gain, meaning that the refinement stage corrects more predictions than it overturns incorrectly.

Static Question Answering shows a different behaviour. Although it achieves the highest hateful-class recall, it consistently reduces overall accuracy and produces negative net correction gains, suggesting increased over-prediction of hate speech. Dynamic Question Answering and Chain-of-Thought reasoning remain much closer to the initial baseline and do not consistently improve overall perfor-

mance.

Figure 3 shows the effect of the confidence threshold on Accuracy. Atomic Fact Extraction remains the strongest strategy across all evaluated thresholds and benefits from more aggressive refinement, while the remaining strategies show little or inconsistent improvement as the refinement rate increases. A qualitative refinement example is provided in Appendix C, while full ablation results are provided in Appendix D.

7 Discussion

Our results show that confidence-gated refinement can modestly improve zero-shot hateful meme detection when the refinement strategy provides useful structure. Atomic Fact Extraction performs best, suggesting that refinement helps most when it guides the model toward relevant evidence rather than simply adding more reasoning steps.

Model comparison: The initial VLM comparison reveals different performance trade-offs across model families. Qwen3-VL-8B achieves the highest overall Accuracy and is therefore selected for refinement, while InternVL3-8B achieves the best Macro F1, hateful-class F1, and hateful-class recall. This suggests that Qwen3-VL-8B is slightly better under our accuracy-based selection criterion, but InternVL3-8B provides stronger hateful-class detection and a more balanced class-wise profile. Future work could therefore apply the same confidence-gated refinement pipeline to InternVL3-8B to test whether its stronger hateful-class behaviour leads to better refined performance.

Refinement strategies: Atomic Fact Extraction performs best because it forces the model to separate observable facts from judgement. This matters for hateful memes, where the final label often depends on whether the image and text jointly target a protected group or express a harmful stereotype. In contrast, Static Question Answering increases hateful-class recall but reduces overall accuracy, suggesting that repeatedly asking about hostility may push the model toward over-predicting hate speech. Dynamic Question Answering and Chain-of-Thought reasoning remain closer to the initial baseline, showing that open-ended reasoning does not automatically improve multimodal hate detection.

Confidence thresholds: The threshold results show that more refinement is not always better.

At the best-performing threshold $t = 0.95$, only 21.77% of examples are routed to refinement, while the remaining predictions are accepted directly. For the weaker refinement strategies, additional reasoning often changes predictions without improving performance. The confidence gate therefore serves two roles: reducing unnecessary computation and preventing reasoning steps that introduce new errors.

Model behaviour: The qualitative examples show that VLMs can be confidently correct on direct cases, but also confidently wrong on subtle or ambiguous memes. Some high-confidence errors occur when the model misses implicit protected-group targeting, while others occur when it confuses offensive or negatively framed content with hate speech. This supports the need to evaluate both accuracy and reasoning behaviour rather than relying only on confidence scores.

Zero-shot and real-world use: The proposed setup does not require task-specific fine-tuning or additional labelled training data, which is useful when meme formats, targets, and coded language change quickly in real-world moderation settings. At the same time, zero-shot performance may remain below supervised systems trained directly on labelled hateful meme data. The remaining high-confidence errors also show that the pipeline should support human moderators by prioritising uncertain cases for review, rather than replacing human judgement in sensitive moderation decisions.

8 Conclusion

This work examined whether confidence-gated structured reasoning can improve zero-shot hateful meme detection without task-specific fine-tuning. We compared four refinement strategies and found that Atomic Fact Extraction performed best, improving both accuracy and Macro F1 over the direct prompting baseline.

Selective refinement helps, but only when the strategy is well-suited to the task. Atomic Fact Extraction works because it separates observable evidence from judgement. Strategies that skipped this step either changed too many predictions or pushed the model toward over-predicting hate speech.

Future work should test the pipeline on larger and multilingual datasets, explore better confidence calibration, and investigate whether the reasoning strategy itself can be chosen adaptively based on the model’s uncertainty.

Limitations

Our experiments use a 3000-example subset of the Hateful Memes benchmark due to the cost of evaluating multiple refinement strategies and confidence thresholds. Results may therefore differ on the full benchmark or larger-scale evaluations.

The study is limited to English memes from a single dataset. Since hateful content depends heavily on cultural context, language, and evolving online references, the findings may not generalise to other domains or multilingual settings.

Our confidence score is based on normalised label likelihood rather than full calibration, and the refinement stage remains sensitive to prompt design and model-specific reasoning behaviour.

Finally, we evaluate recent open-source VLMs in a strict zero-shot setting without fine-tuning. Future work could test larger models, stronger calibration methods, and adaptive refinement strategies on broader datasets.

Ethical Considerations

This project studies hateful meme detection using a dataset that contains offensive and discriminatory content. The examples shown in the paper are included only to illustrate the task and the challenges of multimodal hate detection. They should not be interpreted as endorsement of the views expressed in the memes.

Automated hateful meme detection systems can produce both false positives and false negatives. False positives may incorrectly flag satire, benign humour, or legitimate discussion, while false negatives may fail to detect harmful content. These challenges are particularly difficult in multimodal settings where meaning depends heavily on cultural context, stereotypes, sarcasm, and image-text interaction.

The proposed pipeline also relies on large vision-language models that may inherit biases from their pretraining data. In addition, the refinement stage generates free-form reasoning, which may contain hallucinated or incorrect interpretations. Although confidence-gated refinement reduces unnecessary reasoning on high-confidence examples, it does not guarantee reliable or unbiased predictions.

For these reasons, the proposed system should be viewed as an experimental research pipeline rather than a fully automated moderation system. Real-world deployment would require broader evaluation across languages and communities together

with human oversight and careful error analysis.

Reflection

Work Division The project was completed collaboratively. Nallathambi and Deepak Somesh primarily worked on the initial prediction stage, including model selection experiments, confidence estimation, and baseline evaluation. Divyanshi Singh and Benard Wanyande primarily worked on the refinement stage, including refinement strategy design, prompt experimentation, and refinement analysis. All group members contributed to report writing, presentation preparation, and experimental discussion.

Use of AI Assistants We used AI assistants, including ChatGPT and Claude, during different stages of the project. The assistants were mainly used for understanding related work, summarising papers, and generating LaTeX formatting suggestions.

We found the assistants particularly useful for debugging implementation issues, summarising related work, and improving LaTeX formatting. However, we observed occasional incorrect implementation suggestions and over-interpreted explanations during multimodal reasoning discussions. Because of this, all generated suggestions and outputs were manually verified before being incorporated into the project.

References

- Raúl Gómez, Jaume Gibert, Lluís Gómez, and Dimosthenis Karatzas. 2020. Exploring hate speech detection in multimodal publications. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops*.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330.
- Y. Jang, W. S. Choi, M. Jung, M. Lee, and Byoung-Tak Zhang. 2025. Confidence-guided refinement reasoning for zero-shot question answering. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 6933–6950, Suzhou, China.
- Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The hateful memes challenge: Detecting hate speech in multimodal memes. In *Advances in Neural Information Processing Systems*.

Bo Li, Yuanhan Zhang, Ziyu Liu, and et al. 2024. LLaVA-OneVision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. In *Advances in Neural Information Processing Systems*.

P. Nandi, S. Sharma, and Tanmoy Chakraborty. 2024. **SAFE-MEME: Structured reasoning framework for robust hate speech detection in memes.** *Preprint*, arXiv:2412.20541.

Qwen Team. 2025. **Qwen2.5-VL technical report.** *Preprint*, arXiv:2502.13923.

Joe Stacey, Pasquale Minervini, Haim Dubossarsky, Oana-Maria Camburu, and Marek Rei. 2024. Atomic inference for natural language inference with generated facts as atoms. *arXiv preprint arXiv:2405.06988*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Yuki Yamagishi. 2024. YYama@multimodal hate speech event detection 2024: Simpler prompts, better results. In *Proceedings of the 7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text*, pages 60–66.

Jinguo Zhu, Xiaokang Chen, and et al. 2025. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*.

Ron Zhu, Di Chen, Ning Xiang, and Mengdan Zhou. 2022. Multimodal zero-shot hateful meme detection. *IEEE Access*, 10:96484–96494.

A Correct High-Confidence Initial Predictions

The examples in this appendix are taken from the initial prediction stage of Qwen3-VL-8B.

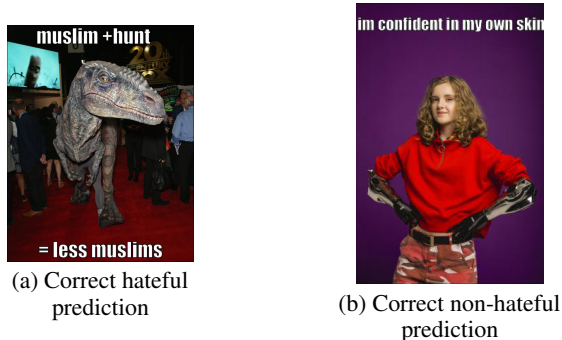


Figure 4: Correct high-confidence initial predictions.

Left example: Ground Truth: Hateful; Prediction: Hateful; Confidence Score: 1.00.

Model Reasoning: “Yes. The image uses the term ‘muslim + hunt = less muslims’ in a context that implies violence or extermination against Muslims, which constitutes hate speech by promoting harm or discrimination against a religious group.”

Right example: Ground Truth: Non-hateful; Prediction: Non-hateful; Confidence Score: 0.9998.

Model Reasoning: “No. The text ‘im confident in my own skin’ promotes self-acceptance and body positivity, which are positive messages. The image shows a person with prosthetic limbs, and the phrase reinforces confidence regardless of physical appearance, which is empowering and not hateful.”

Analysis: These examples show that the direct zero-shot prediction stage can correctly classify both explicit hateful content and clearly non-hateful supportive content with very high confidence.

B Wrong High-Confidence Initial Predictions



(a) Missed implicit hateful context



(b) Non-hateful meme misclassified as hateful

Figure 5: Wrong high-confidence initial predictions.

Left example: Ground Truth: Hateful; Prediction: Non-hateful; Confidence Score: 0.9579.

Model Reasoning: “No. The text expresses disappointment or frustration about a child not being invited to a nativity play, which is a common social or cultural event. It does not contain hate speech, as it does not target individuals or groups based on race, religion, ethnicity, gender, or other protected characteristics.”

Right example: Ground Truth: Non-hateful; Prediction: Hateful; Confidence Score: 0.9975.

Model Reasoning: “Yes. The image contains hate speech because it depicts a disturbing and dehumanizing scenario involving animals, combined

Rank	Strategy	Thr.	Acc.	Macro F1	H-F1	H-Rec.	Ref.	Rate	Changed	Gain
1	Atomic	0.95	0.7230	0.7109	0.6519	0.7312	653	0.2177	60	16
2	Atomic	0.75	0.7223	0.7097	0.6493	0.7246	426	0.1420	48	14
3	Atomic	0.85	0.7220	0.7096	0.6496	0.7265	495	0.1650	53	13
4	Atomic	0.60	0.7220	0.7093	0.6484	0.7227	371	0.1237	45	13
5	CoT	0.60	0.7187	0.7049	0.6412	0.7086	371	0.1237	25	3
6	CoT	0.75	0.7183	0.7046	0.6409	0.7086	426	0.1420	26	2
7	CoT	0.85	0.7177	0.7040	0.6403	0.7086	495	0.1650	28	0
8	Dynamic QA	0.85	0.7173	0.7063	0.6493	0.7378	495	0.1650	91	-1
9	CoT	0.95	0.7173	0.7037	0.6401	0.7086	653	0.2177	29	-1
10	Dynamic QA	0.75	0.7170	0.7056	0.6476	0.7331	426	0.1420	82	-2
11	Dynamic QA	0.60	0.7163	0.7048	0.6464	0.7312	371	0.1237	80	-4
12	Static QA	0.75	0.7150	0.7070	0.6587	0.7754	426	0.1420	178	-8
13	Static QA	0.60	0.7150	0.7066	0.6570	0.7697	371	0.1237	166	-8
14	Dynamic QA	0.95	0.7150	0.7044	0.6483	0.7406	653	0.2177	104	-8
15	Static QA	0.85	0.7133	0.7060	0.6595	0.7829	495	0.1650	199	-13
16	Static QA	0.95	0.7120	0.7056	0.6622	0.7961	653	0.2177	231	-17

Table 3: Full confidence-gated refinement ablation results across strategies and thresholds. H-F1 and H-Rec. denote hateful-class F1 and hateful-class recall. Ref. denotes refined examples, Rate denotes refinement rate, Changed denotes changed predictions, and Gain denotes net correction gain over initial prediction baseline.

with a caption that implies a callous or violent decision-making process. The juxtaposition of the animals and the text promotes cruelty and insensitivity, which can be considered hateful or offensive.”

Analysis: These examples show two types of high-confidence errors. In the left example, the model misses the implicit hateful context suggested by the military imagery and interprets the meme as ordinary social disappointment. In the right example, the model over-predicts hate speech by treating ambiguous or negatively framed content as hateful despite the absence of a protected-group target.

C Incorrect Initial Predictions Corrected by Refinement

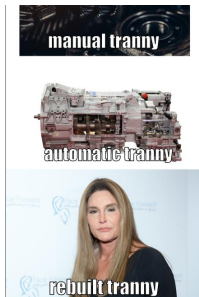


Figure 6: Prediction corrected by Atomic Fact Extraction.

Ground Truth: Hateful; Initial Prediction: Non-hateful; Final Prediction: Hateful; Confidence Score: 0.148; Strategy: Atomic Fact Extraction. For Figure 6

Initial Reasoning: “No. The image uses automotive terminology in a humorous or sarcastic context, likely referencing the complexity or perceived superiority of automatic transmissions.”

Extracted Atomic Facts:

- The top image shows a close-up of mechanical gears labeled “manual tranny.”
- The middle image shows a complex, transparent automatic transmission labeled “automatic tranny.”
- The bottom image shows a photo of Caitlyn Jenner labeled “rebuilt tranny.”

Refined Reasoning: The refinement identifies Caitlyn Jenner as the target and interprets the phrase “rebuilt tranny” as a derogatory metaphor that equates her gender identity with mechanical repair. It therefore classifies the meme as hateful due to its demeaning and mocking portrayal.

Analysis: The initial prediction treats the meme as harmless humour. Atomic Fact Extraction makes the model explicitly identify the target group, the stereotype, and the image-text connection, which helps it recover the correct hateful label.

D Full Refinement Ablation Results

Table 3 reports the complete confidence-gated refinement ablation across all refinement strategies and confidence thresholds.